

Task 1

Explain why you chose Transform Data over Load, and state the three ways to access Use First Row as Headers in Power Query.

Answer

A) I chose **Transform Data** over **load** because it opens Power Query Editor first —use this to clean, rename columns, remove blanks, change data types, etc., before my data enter Power BI

B)

1) Home Ribbon

Home tab → Use First Row as Headers (*fastest method*)

2) Transform Tab

Transform tab → Use First Row as Headers (*alternative location*)

3) Right-click Column

Right-click the 'Column1' header → Use First Row as Headers in the context menu.

Task 2

State where a query with load disabled still appears in Power BI, and describe one scenario where disabling load is useful.

Answer

A)

1) Power Query Editor

When load is **Disabled**, the query runs but its output is not visible in the Fields pane — it only serves as a staging step. It shows with an *italic* name to indicate load is disabled

B) Disabling Load is Useful because;

To **improve performance** — fewer loaded tables result faster model

Task 3

List every column that contains errors and its approximate error percentage.

Answer

A) Column Quality =0%

B) Column Profile =0%

Task 4

Filter the Quantity column to show only Error rows and record how many exist.

Answer

A) Quality Column Error = 70

Task 5

Explain the difference between Replace Errors and Remove Errors, and state which one keeps the row in the dataset.

Answer

Replace Errors Swaps the 'Error' cell with a value I specify —1. The row is KEPT, only the error cell is replaced. While,

Remove Errors Permanently deletes entire rows where the selected column contains an error. Use when rows with errors are unusable and should not be in the model.

Hence, Replace Errors keep the Row in the Dataset

Task 6

Use Column Quality to find the percentage of null values in the Discount column

Answer

Column Quality =19%

Task 7

Filter the OrderDate column to exclude all null values and any invalid text entries such as ERROR_DATE, #VALUE! and BAD-DATE and state whether this permanently deletes those rows from the source file, and explain what happens on the next data refresh.

Answer

- A) Filtering the order column to exclude all null values did not permanently delete those rows from the source file
- B) On the next refresh, all the null values such as ERROR_DATE, #VALUE! and BAD-DATE are excluded and the column has no empty row.

Task 8

Assign the Correct Data Type to Each Column Apply the correct data types in Power Query and complete this table

Column	Correct Data Type	Reason
OrderID	Text	Because it is a unique identifier
OrderDate	Date	Because it Stores date values
CustomerID	Text	Because it is a unique identifier
Quantity	Whole Number	Because it Represents countable items
UnitPrice	Decimal Number	Because it Represents monetary values
Discount	Decimal Number	Because Discount is typically expressed as a decimal or Percentage

Task 9

- A) Enable Column Distribution on Order ID and record the Distinct and Unique counts and explain what the difference between them means
- B) Remove duplicate rows based on Order ID, keeping the first occurrence. Record the row count before and after

Answer

A) Column Distribution

- Distinct = 860
- Unique = 722

A₂) **DISTINCT** Returns all values that appear at least once — removes duplicates but keeps one copy of each repeated value. While,

UNIQUE Returns only values that appear exactly once — completely removes any value that appears more than once.

B)

Row Count Before = 3071

Row Count After = 1687

Task 10

Remove all rows where every cell is blank and explain the difference between this operation and removing nulls in a specific column

Answer

Remove Blank Row — it only removes rows that have absolutely no useful data anywhere. While,

Remove Null — it enforces that a particular column must always have a value, even if it means losing rows with partial data

Task 11

State whether Trim removes spaces between words and explain what it actually removes.

Answer

- A) No — Trim does NOT remove spaces between word
- B) Trim only removes leading and trailing spaces — spaces that appear at the beginning and end of a text value.

Task 12

Record how many distinct state values remain after cleaning

Answer

Distinct = 12

Task 13

State what step name Power Query records for this action, and explain how you would handle names that split incorrectly due to a middle name

Answer

- A) Split Column by delimiter
- B) Due to Middle name Instead of splitting at left-most only, split into multiple columns (Each of Occurrence), or trim the column first then split by delimiter to left-most only

Task 14

list the unique values in Order Year, and explain what would have happened to the original column if the Transform tab had been used instead of Add Column

Answer

- A) 2023
- B) If **Transform Tab** is used, it will result to Destructive (works on the original column and replaces its values entirely). While,

Add Column will result to Non-destructive — works beside the original column leaving it untouched

Task 15

Record the total row count of sales_full_2023 and explain what happens when a column exists in one table but not the other during an Append

Answer

- A) Total Rows = 1612
- B) Unmatched columns return null values during an Append

Task 16

Explain why a Left Outer join is used here and what rows would be lost with an Inner join instead

Answer

- A) Left Outer join is used because I want to return all rows from LEFT table (Sales -q2), matching from Right
- B) Only rows that MATCH in both tables (Sales Data and Sales-q2) will be return if Inner join is used. Hence, those unmatched row would be lost.

Task 17

Open the Query Dependencies diagram, and describe what the diagram shows

Answer

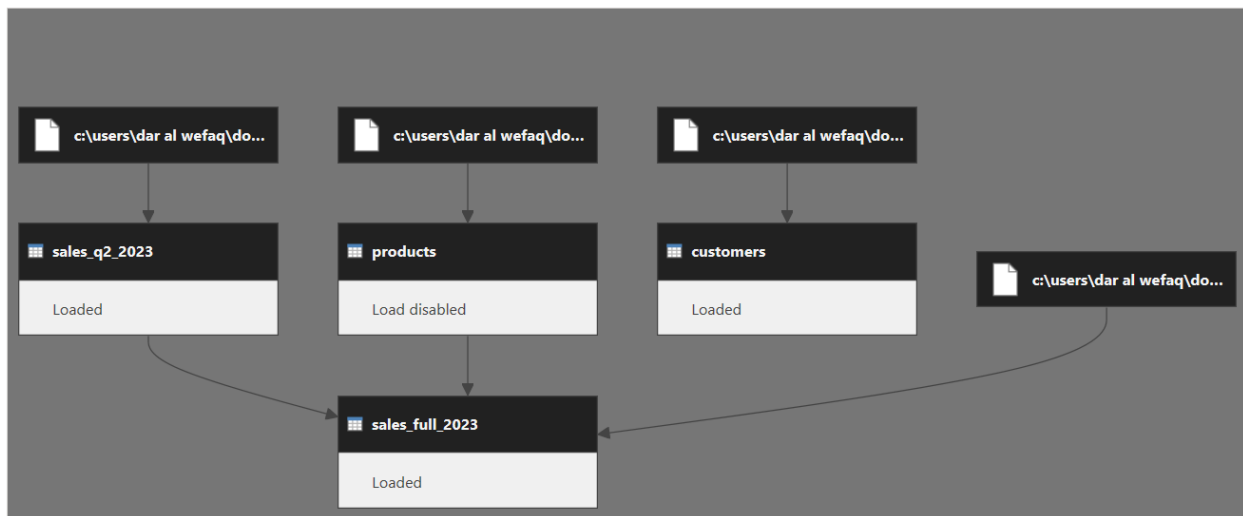
- A) All the four tables are connected to the source
- B) All the tables are loaded except the Product table (it was Disable)
- C) Sales Full- 2023 References both Product base Query and Sales data Source.

Order ID	Order Date	Customer ID	Product ID	ProductName	Category	SubCategory	Quantity	Unit Price	Discount	State	Region
TXN-ORD10001	6/27/2023	C0031	P013	Nike Air Max Running Shoe	FASHION	Footwear	19	43564	0.15	Port Harcourt	South South
TXN-ORD10002	1/30/2023	C0214	P012	Binatone Microwave 20L	APPLIANCES	Kitchen Appliances	1	45258	0.2	Benue	North Central
TXN-ORD10003	1/3/2023	C0044	P020	Canon PIXMA Printer	ELECTRONICS	Printers	14	89296	0	Katsina	North West
TXN-ORD10004	1/22/2023	C0163	P004	HP Pavilion Laptop	ELECTRONICS	Laptops	12	374839	0.05	Ogun	South West
TXN-ORD10005	3/25/2023	C0244	P016	Indomie Pack of 40	FMCG	Food	22	15617	0.1	Lagos	South West
TXN-ORD10006	7/28/2023	C0223	P010	Nexus Air Conditioner 1.5H	APPLIANCES	Air Conditioners	17	206425	0.2	Ogun	South West
TXN-ORD10007	9/23/2023	C0160	P005	Dell Inspiron 15	ELECTRONICS	Laptops	15	332563	0.05	Enugu	South East
TXN-ORD10008	8/31/2023	C0100	P015	Ankara Wax Fabric 5m	FASHION	Fabrics	18	9503	0.15	Anambra	South East
TXN-ORD10009	2/2/2023	C0169	P020	Canon PIXMA Printer	ELECTRONICS	Printers	5	91497	0.15	Ibadan	South West
TXN-ORD10010	2/18/2023	C0085	P005	Dell Inspiron 15	ELECTRONICS	Laptops	22	351969	0	Ondo	South West
TXN-ORD10011	3/4/2023	C0036	P005	Dell Inspiron 15	ELECTRONICS	Laptops	25	347253	0	Ibadan	South West
TXN-ORD10012	7/15/2023	C0192	P002	iPhone 14	ELECTRONICS	Smartphones	3	528410	0.1	Ibadan	South West
TXN-ORD10013	7/18/2023	C0219	P008	Haier Thermocool Fridge	APPLIANCES	Refrigerators	5	154927	0.15	Kano	North West
TXN-ORD10014	9/29/2023	C0219	P013	Nike Air Max Running Shoe	FASHION	Footwear	3	42341	0	Kano	North West
TXN-ORD10015	2/27/2023	C0239	P001	Samsung Galaxy A54	ELECTRONICS	Smartphones	13	210923	0	Ibadan	South West

Sales Rep	Segment	First Name	Last Name	Order Year	Order Month No	Order Month Name	Total Sales	Net Sales	Sales Tier	Cost Price	Profit
Fatima Sule	Home Office	Tobi	Nwofor	2023	6	June	827716	703558.6	High	25000	802716
Ngozi Okoro	Corporate	Ibrahim	Ezeani	2023	1	January	45258	36206.4	Low	35000	10258
Tunde Bello	Consumer	Fatima	Ogundele	2023	1	January	1250144	1250144	High	65000	1185144
Kemi Adeyemi	Consumer	Damilola	Sule	2023	1	January	4498068	4273164.6	High	320000	4178068
Ngozi Okoro	Consumer	Tochukwu	Bello	2023	3	March	343574	309216.6	Medium	3500	340074
Emeka Chukwu	Home Office	Kelechi	Taiwo	2023	7	July	3509225	2807380	High	175000	3334225
Musa Ibrahim	Corporate	Uche	Adeyemi	2023	9	September	4988445	4739022.75	High	280000	4708445
Chioma Nwosu	Consumer	Bola	Olawale	2023	8	August	171054	145395.9	Medium	8000	163054
Dare Lawal	Home Office	Ngozi	Chukwu	2023	2	February	457485	388862.25	Medium	65000	392485
Tunde Bello	Home Office	Eze	Nwachukwu	2023	2	February	7743318	6968986.2	High	280000	7463318
Tobi Adeleke	Home Office	Tunde	Onyeka	2023	3	March	8681325	8681325	High	280000	8401325
Dare Lawal	Consumer	Stella	Obiora	2023	7	July	1585230	1426707	High	450000	1135230
Tobi Adeleke	Consumer	Bola	Okafor	2023	7	July	774635	658439.75	High	125000	649635

Data View showing the final table with all new columns

Query Dependencies



Query Dependencies diagram